

Lab 6: IoT

Course: CPE371 Big Data Engineering

Group Members:

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Overview

#	step	short detail
0	<u>Facts checked first</u>	versions and data shape, verified not assumed
0	<u>Requirements coverage</u>	every numbered instruction mapped to where it is met
1	<u>Step 1 - Environment</u>	JDK 17, virtual environment, dependencies
2	<u>Step 2 - Publish and land</u>	CSV replayed over MQTT into the landing zone
3	<u>Step 3 - Windowed averages</u>	the four figures the activity asks for
4	<u>Step 4 - Watermark and alerts</u>	late-data eviction, ALERT/OK status
5	<u>Step 5 - Assignment 1</u>	dweet.cc loop, read-back, ThingSpeak fields
6	<u>Step 6 - Assignment 2</u>	/CPE_HOUSE/ topics, publisher + subscriber
7	<u>Step 7 - Actuators</u>	four control rules, LED commands
8	<u>Step 8 - End to end</u>	all four stages at once, measured

Facts checked before starting

Thing	Value
Dataset	1088 rows each row emits one indoor and one outdoor event, so 2176 events per replay
CSV encoding	UTF-8 with BOM - encoding= 'utf-8-sig' is required or the first column name breaks
CSV columns	room_temperature, room_humidity, outdoor_temperature, outdoor_humidity, location_latitude, location_longitude, AQI, AirCondition_Status
Outdoor temperatures	45–48 °C in the source data, above the 35 °C alert threshold, so Outdoor windows are permanently ALERT
Python	3.14.6 - PySpark 4.2 lists 3.10–3.14 as supported
Java	Temurin 17.0.20.1; Spark 4.x requires JDK 17 or 21
PySpark	4.2.0
paho-mqtt	2.1.0 - 2.x needs an explicit CallbackAPIVersion
Broker	broker.mqtttdashboard.com:1883, public and shared with the whole class
Floorplan nodes	5 rooms with sensors, 10 with LED controllers; WINE CELLAR has no sensor though the cellar rule needs one

Two consequences to remember:

1. The project README claimed Python 3.9–3.11 and warned that Java 21+ breaks PySpark. Both applied to Spark 3.x and are wrong for Spark 4.2.
 2. /CPE_HOUSE/# returns other groups' topics because the broker is shared. That is expected, and it is the argument for a unique topic root in a real deployment.
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Lab requirements coverage

Activity, from the lab slide (question.png):

#	Requirement	Met by	Status
1	Publish the provided sensor data to the MQTT broker	mqtt_publisher.py, step 2	done
2	Subscribe to the topic and write each event to the JSON landing directory	mqtt_bridge.py, step 2	done
3	Process newly arriving files with Spark Structured Streaming	spark_streaming.py readStream, step 3	done
4	Windowed averages (10 s) for indoor/outdoor temperature and humidity	zone-level groupBy, step 3	done
5	Display the updated results	console sink, batch output in step 3	done

Assignment 1, from assignment1.ipynb:

#	Requirement	Met by	Status
1	Simulate the IoT node by reading the datalog	csv.DictReader with utf-8-sig, cell 4	done
2	POST the HTTP message to dweet.cc in a loop every 5 or 10 seconds	cell 7, 5 s interval, form-encoded	done
3	Create a ThingSpeak channel and chart the data stored at dweet.cc , using as many fields as possible	cell 9 posts to dweet, reads it back, forwards that content to fields 1–8	done
4	Capture the dashboard screenshot and submit with the file	see Dashboard screenshot	done

Assignment 2, from assignment2.ipynb:

#	Requirement	Met by	Status
1	Split into publisher and subscriber teams	two notebooks, one role each	done
2	Publisher: loop-publish all topics to <code>broker.mqtttdashboard.com</code> every 5–10 s, varying the values	3 cycles at 5 s, random walk per room	done
3	Subscriber: subscribe and display, then practise changing the topic	<code>#</code> , <code>+/temperature</code> and <code>LIVING/#</code> all demonstrated	done
4	Submit publisher and subscriber as two separate .ipynb files, keeping the run output	both executed, outputs retained	done

Requirement A1.3 says the dashboard must use *the data stored at dweet.cc*. Each iteration therefore posts a row to dweet, reads it back with GET

`/get/latest/dweet/for/<thing>`, and sends that read-back content to ThingSpeak - not a second pass over the CSV.

Step 1 - Environment

1. Install JDK 17 and point JAVA_HOME at it.

```
export JAVA_HOME=~/.jdk/jdk-17.0.20.1+1/Contents/Home
export PATH="$JAVA_HOME/bin:$PATH"
java -version # expect 17.x
```

2. Create the virtual environment and install dependencies.

```
cd ~/Documents/KMUTT/CPE/3.1/CPE371/Lab/Lab5
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

3. Confirm Spark actually starts before going further; this is the step that fails if Java is wrong.

```
python -c "
from pyspark.sql import SparkSession
s = SparkSession.builder.master('local[*]').getOrCreate()
print('Spark', s.version); s.stop()"
```

Step 2 - Publish and land

Activity steps 1 and 2. Two terminals, bridge first - start the publisher first and its early events go to nobody.

```
# terminal 1
python src/mqtt_bridge.py

# terminal 2, once the bridge prints "Subscribed to topic hierarchy"
python src/mqtt_publisher.py --interval 0.5 --max-records 40 --mode both
```

Check what landed:

```
ls iot_landing/*.json | wc -l
cat "$(ls iot_landing/*.json | head -1)"
```

The bridge writes `.{event_id}.tmp` then renames it to `{event_id}.json`. The leading dot stops Spark reading a half-written file, and the rename is atomic.

Step 3 - Windowed averages

Activity steps 3, 4 and 5.

```
python src/spark_streaming.py --trigger "5 seconds" --output-mode update
```

Grouping is by window and spatial zone, which yields exactly the four figures the activity asks for. Add `--by-room` for a per-room breakdown.

Activity metric	Column
Indoor temperature	avg_temperature where spatial_zone = Indoor
Indoor humidity	avg_humidity where spatial_zone = Indoor
Outdoor temperature	avg_temperature where spatial_zone = Outdoor
Outdoor humidity	avg_humidity where spatial_zone = Outdoor

Real console output, batch 5 of the recorded run:

```
Shell
-----
Batch: 5
-----
|window
|spatial_zone|avg_temperature|avg_humidity|avg_aqi|max_temperature|min_temperatur
e|event_count|status|
|{2026-09-14 20:18:20, 2026-09-14 20:18:30}|Outdoor      |45.37          |29.99
    |69.0    |46.3          |44.9          |9          |ALERT |
|{2026-09-14 20:18:20, 2026-09-14 20:18:30}|Indoor        |26.04          |33.9
    |69.0    |26.6          |25.2          |9          |OK    |
```

Step 4 - Watermark and alerts

Setting	Value	Effect
window / slide	10 s	tumbling
watermark	30 s	evicts closed-window state, drops later arrivals
temperature alert	> 35.0 °C	status becomes ALERT
humidity alert	> 70.0 %	status becomes ALERT
checkpoint	checkpoints/spark_iot_metrics	restart resumes state

Verified behaviour:

1. A late event aimed at an already-closed window was dropped - the window's `event_count` stayed at 3 and its average stayed 22.0, instead of becoming 4 and 41.25.
2. Restarting against the same checkpoint resumed state rather than reprocessing: the window went 2 → 4 events with the average moving 21.0 → 23.0.
3. Boundaries are strict >: 35.0 °C gives OK, 35.1 °C gives ALERT; 70.0 % gives OK, 70.1 % gives ALERT.

--output-mode completely retains every aggregate ever computed, so the watermark cannot evict anything and memory grows. Use update.

Step 5 - Assignment 1: dweet.cc and ThingSpeak

1. Post the datalog to dweet.cc every 5 seconds, then read it back to prove it arrived.

```
python src/cloud_forwarder.py --count 1
```

2. dweet.cc requires **form encoding**. Posting JSON returns HTTP 200 with a failure body:

```
curl -s -X POST -H 'Content-Type: application/json' -d '{"t":1}' \
https://dweet.cc/dweet/for/cpe371-67070503467
# {"this": "failed", "because": "No data provided"}
```

3. Channel field mapping - the assignment prefers as many fields as possible, so all eight are used.

Field	Meaning	Field	Meaning
field1	Indoor temperature	field5	AQI
field2	Indoor humidity	field6	AirCondition_Status
field3	Outdoor temperature	field7	Latitude
field4	Outdoor humidity	field8	Longitude

4. Feed ThingSpeak from the dweet store, as requirement A1.3 specifies.

```
export THINGSPEAK_WRITE_KEY=<your-16-char-write-key> # never commit the realShell key
python src/cloud_forwarder.py --count 1 --from-dweet
```

The free tier refuses updates less than 15 seconds apart and answers 0 when it does - that rate limit is the usual reason a channel stays empty.

Live read-back from the recorded run:

[+] Sweet read-back OK (8 fields): room_temperature 26.8, room_humidity 35.9,
outdoor_temperature 45.8, outdoor_humidity 25.2, AQI 68, AirCondition_Status 1,
location_latitude 13.85548736, location_longitude 100.5700254

[*] Forwarding 8 fields from dweet.cc -> ThingSpeak

Dashboard screenshot

CPE371-Lab4

Channel ID: 3493199
Author: mwa0000042381373
Access: Private

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

+ Add Visualizations

+ Add Widgets

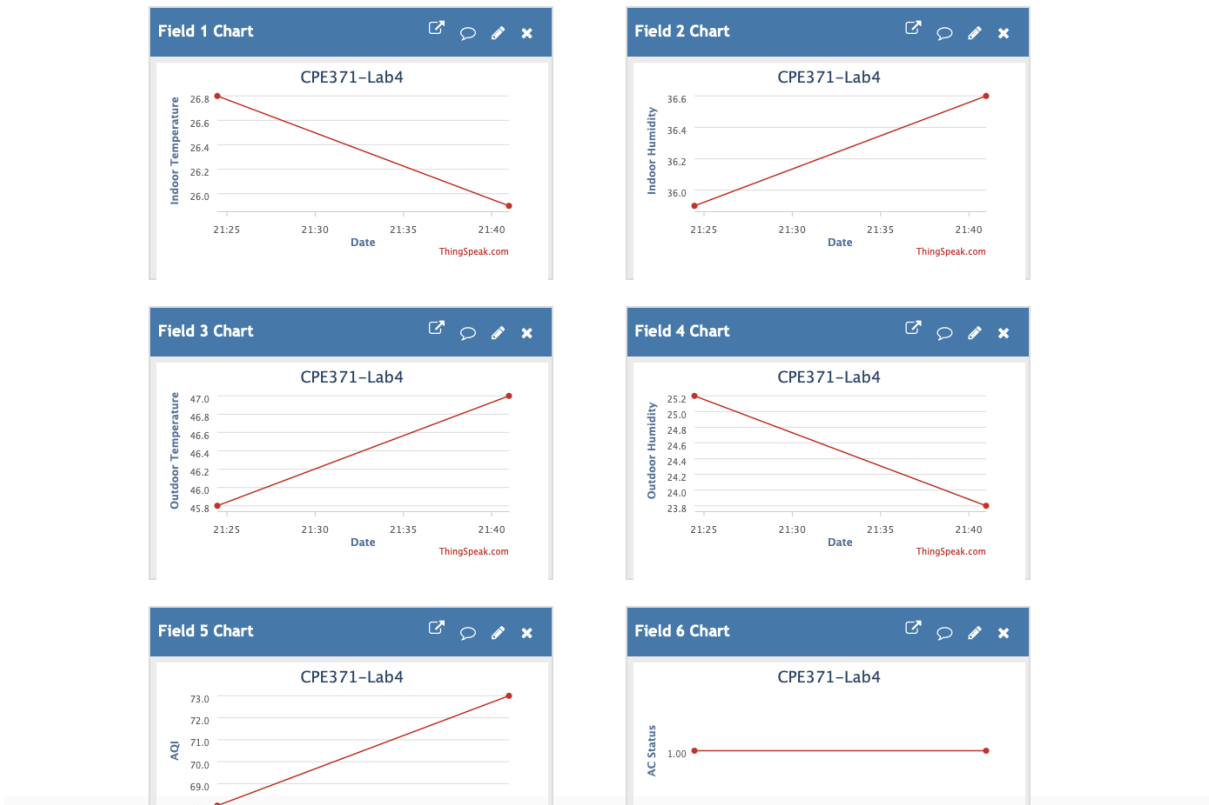
Export recent data

MATLAB Analysis

MATLAB Visualization

Channel Stats

Created: 30 minutes ago
Entries: 8



Channel CPE371-Lab4, ID 3493199, 8 entries. Fields 1–6 are visible with their labels; fields 7–8 (latitude, longitude) are below the fold in this capture.

Step 6 - Assignment 2: CPE_HOUSE topics

Topic scheme, as the handout specifies:

/CPE_HOUSE/<ROOM>/temperature

/CPE_HOUSE/<ROOM>/humidity

/CPE_HOUSE/<ROOM>/LED/status reported state

/CPE_HOUSE/<ROOM>/LED/set commanded state

Rooms	Nodes	Topics per cycle
LAUNDRY, LIVING, KITCHEN, TERRACE, GYM	sensor + LED	3 each
DINING, GARAGE, BATH, TERRACE_POOL, PATIO	LED only	1 each
Total	10 rooms	20

Results:

1. Publisher ran 3 cycles at 5-second intervals, 20 messages per cycle, retain=True.
2. Subscriber on /CPE_HOUSE/# received 156 topics; /CPE_HOUSE/+ /temperature narrowed to 18 and /CPE_HOUSE/LIVING/# to 5.
3. The set → status round trip was acknowledged in both directions, ON and OFF.

Because the publisher retains messages, the subscriber receives the last value of every topic the moment it subscribes - the two notebooks do not have to run at the same time.

Subscriber output

```
(.venv) sakura@MacBookAirPiti-2 Lab5 % cd ~/Documents/KMUTT/CPE/3.1/CPE371/Lab/Lab5 && source .venv
/bin/activate
python -c "
import time, paho.mqtt.client as mqtt
got={}
def oc(c,u,f,rc,p=None): c.subscribe('/CPE_HOUSE/#')
def om(c,u,m): got[m.topic]=m.payload.decode('utf-8','replace')
c=mqtt.Client(mqtt.CallbackAPIVersion.VERSION2); c.on_connect=oc; c.on_message=om
c.connect('broker.mqttdashboard.com',1883,45); c.loop_start(); time.sleep(6); c.loop_stop()
for t in sorted(got): print(f'{t:46s} {got[t][:60]}')"
```

/CPE_HOUSE/34434665/BATH/LED/status	OFF
/CPE_HOUSE/34434665/DINING/LED/status	OFF
/CPE_HOUSE/34434665/GARAGE/LED/status	OFF
/CPE_HOUSE/34434665/GUEST_BEDROOM/humidity	57.9
/CPE_HOUSE/34434665/GUEST_BEDROOM/temperature	27.4
/CPE_HOUSE/34434665/GYM/LED/status	OFF
/CPE_HOUSE/34434665/GYM/humidity	62.5
/CPE_HOUSE/34434665/GYM/temperature	26.3
/CPE_HOUSE/34434665/KITCHEN/LED/status	OFF
/CPE_HOUSE/34434665/KITCHEN/humidity	61.1
/CPE_HOUSE/34434665/KITCHEN/temperature	27.9
/CPE_HOUSE/34434665/LAUNDRY/LED/status	OFF
/CPE_HOUSE/34434665/LAUNDRY/humidity	59.6
/CPE_HOUSE/34434665/LAUNDRY/temperature	27.0
/CPE_HOUSE/34434665/LIVING/LED/status	OFF
/CPE_HOUSE/34434665/LIVING/humidity	59.7
/CPE_HOUSE/34434665/LIVING/temperature	26.7
/CPE_HOUSE/34434665/PATIO/LED/status	OFF
/CPE_HOUSE/34434665/TERRACE_LOWER/LED/status	OFF
/CPE_HOUSE/34434665/TERRACE_UPPER/LED/status	OFF
/CPE_HOUSE/34434665/TERRACE_UPPER/humidity	59.2
/CPE_HOUSE/34434665/TERRACE_UPPER/temperature	27.4
/CPE_HOUSE/34434665/WINE_CELLAR/humidity	59.5
/CPE_HOUSE/34434665/WINE_CELLAR/temperature	27.2
/CPE_HOUSE/67070503446/BATH/LED/status	OFF
/CPE_HOUSE/67070503446/DINING/LED/status	OFF
/CPE_HOUSE/67070503446/GARAGE/LED/status	OFF
/CPE_HOUSE/67070503446/GUEST_BEDROOM/humidity	59.7
/CPE_HOUSE/67070503446/GUEST_BEDROOM/temperature	26.9
/CPE_HOUSE/67070503446/GYM/LED/status	OFF
/CPE_HOUSE/67070503446/GYM/humidity	59.3
/CPE_HOUSE/67070503446/GYM/temperature	26.5
/CPE_HOUSE/67070503446/KITCHEN/LED/status	OFF

The capture shows `/CPE_HOUSE/#` returning other groups' topics, which are prefixed with a student ID (`/CPE_HOUSE/34434665/...`, `/CPE_HOUSE/67070503446/...`). Ours carry no prefix, so they sort after the numeric ones and fall below the fold here.

That prefix difference matters. This project publishes to the handout's literal example path, `/CPE_HOUSE/<ROOM>/temperature`, and at least one other group does the same - 35 unprefix topics are live but only 20 of them are ours, and topics such as `/CPE_HOUSE/BATH/data` are not published by this code at all. The topic root is therefore not unique enough to prove ownership on a shared broker. Adding a student-ID level, as the other groups did, would fix it.

Step 7 - Actuator rules

Rule	Condition	Action
Air conditioner	room == living and temperature > 28.0 °C	POWER_ON
Wine cellar	temperature outside 12.0–18.0 °C	COOL or HEAT
Air purifier	AQI > 100	FAN_HIGH
Alert LED	window status is ALERT	BLINK

```
python src/actuator_logic.py                                # rule demo      Shell
python src/cloud_forwarder.py --count 1 --actuate           # against live
events
python src/cloud_forwarder.py --count 1 --actuate --publish-commands # send
LED commands
```

Rules are room-scoped: a kitchen reading at 35 °C produces no air-conditioner decision, because the rule names the living room.

The wine cellar needs a sensor added to the floorplan before its 12–18 °C rule can act on real readings; it is declared in code with `sensor : False` until then.

Step 8 - End-to-end run

All four stages concurrently, 40 records at 0.5 s.

Metric	Value	Metric	Value
Events published	80	Indoor average temperature	26.29 °C
Events landed	80	Indoor average humidity	34.14 %
Delivery rate	100.0 %	Outdoor average temperature	46.05 °C
Spark micro-batches	8	Outdoor average humidity	28.29 %
Latency median	1.06 s	Latency p95	1.44 s
Actuator decisions	15	Errors in any stage log	0

Windowed state is held at 2–4 rows per batch across the run, so the watermark is evicting closed windows rather than accumulating them. 38 checks pass across the four stages.

CPE_HOUSE topic evidence

```
/CPE_HOUSE/BATH/LED/status      ON
/CPE_HOUSE/BATH/data             {"room": "BATH", "timestamp": "13:58:01", "LED_sta
/CPE_HOUSE/DINING/LED/set        1
/CPE_HOUSE/DINING/LED/status     OFF
/CPE_HOUSE/DINING/data           {"room": "DINING", "timestamp": "13:58:01", "LED_s
/CPE_HOUSE/GARAGE/LED/status     OFF
/CPE_HOUSE/GARAGE/data           {"room": "GARAGE", "timestamp": "13:58:01", "LED_s
/CPE_HOUSE/GYM/LED/set          1
/CPE_HOUSE/GYM/LED/status       ON
/CPE_HOUSE/GYM/data              {"room": "GYM", "timestamp": "13:58:01", "temperat
/CPE_HOUSE/GYM/humidity          52.0
/CPE_HOUSE/GYM/temperature       29.5
/CPE_HOUSE/KITCHEN/LED/set      {"command": 1, "action": "ON", "timestamp": 175855
/CPE_HOUSE/KITCHEN/LED/status   ON
/CPE_HOUSE/KITCHEN/data          {"room": "KITCHEN", "timestamp": "13:58:01", "temp
/CPE_HOUSE/KITCHEN/humidity      55.2
/CPE_HOUSE/KITCHEN/temperature   25.6
/CPE_HOUSE/LAUNDRY/LED/set      ON
/CPE_HOUSE/LAUNDRY/LED/status   ON
/CPE_HOUSE/LAUNDRY/data          {"room": "LAUNDRY", "timestamp": "13:58:01", "temp
/CPE_HOUSE/LAUNDRY/humidity      44.8
/CPE_HOUSE/LAUNDRY/temperature   28.3
/CPE_HOUSE/LIVING/LED/set       OFF
/CPE_HOUSE/LIVING/LED/status    OFF
/CPE_HOUSE/LIVING/data           {"room": "LIVING", "timestamp": "13:58:01", "tempe
/CPE_HOUSE/LIVING/humidity       60.9
/CPE_HOUSE/LIVING/temperature    25.1
/CPE_HOUSE/PATIO/LED/status     OFF
/CPE_HOUSE/PATIO/data            {"room": "PATIO", "timestamp": "13:58:01", "LED_st
/CPE_HOUSE/PATIO/humidity        nan
/CPE_HOUSE/TERRACE/LED/status   ON
/CPE_HOUSE/TERRACE/data          {"room": "TERRACE", "timestamp": "13:58:01", "temp
/CPE_HOUSE/TERRACE/humidity      57.3
/CPE_HOUSE/TERRACE/temperature   30.3
/CPE_HOUSE/TERRACE_POOL/LED/status OFF
```

Live subscriber output filtered to this group's rooms, captured from `broker.mqtttdashboard.com`.

Defects found

Six defects, every one in code already marked complete, every one found by running it.

#	Where	Defect	Effect if unfixed	Status
1	Spark reader	<code>multiLine</code> not set on indented JSON	850 rows parsed, every field NULL	fixed
2	Spark aggregation	grouped by room as well as zone	the activity's four figures never produced	fixed
3	Spark categorization	bed room missing from the indoor list	those events classified Unspecified	fixed
4	Spark timestamps	<code>to_timestamp</code> under ANSI mode	one malformed event aborted the whole query	fixed
5	dweet.cc client	posted JSON instead of form data	every post failed while reporting success	fixed
6	Actuator rules	placeholder thresholds, no room awareness	every rule fired incorrectly	fixed

Defect 4 is the one worth explaining. Spark 4 enables `spark.sql.ansi.enabled` by default, and under ANSI mode `to_timestamp` raises instead of returning NULL. A single malformed event_time therefore killed the stream, and the `coalesce` fallback to `_received_at` could never fire because the exception came first. `try_to_timestamp` restores the intended behaviour.

Defect 5 is the one most likely to pass unnoticed: dweet.cc answers HTTP 200 while rejecting a JSON body, so a status-code check reports success on a post that stored nothing.

Troubleshooting

Problem	Fix
Unable to locate a Java Runtime	JDK not installed or JAVA_HOME unset - see Step 1
Spark rows all NULL	JSON reader needs multiLine=true for indented files
CANNOT_PARSE_TIMESTAMP aborts the query	use try_to_timestamp, not to_timestamp, under Spark 4
This query does not support recovering from checkpoint	grouping changed - rm -rf checkpoints/
Spark memory grows over a long run	--output-mode complete never evicts; use update
No events land in <code>iot_landing/</code>	publisher started before the bridge subscribed
dweet says "this": "failed"	posted as JSON; send form-encoded data instead
ThingSpeak returns 0	updates faster than 15 s apart, or a wrong write key
<code>/CPE_HOUSE/#</code> returns other people's data	shared public broker; expected
Subscriber receives nothing	timer started before the connection completed - wait for <code>on_connect</code>
<code>mqtt.Client()</code> deprecation warning	paho-mqtt 2.x wants <code>CallbackAPIVersion.VERSION2</code>